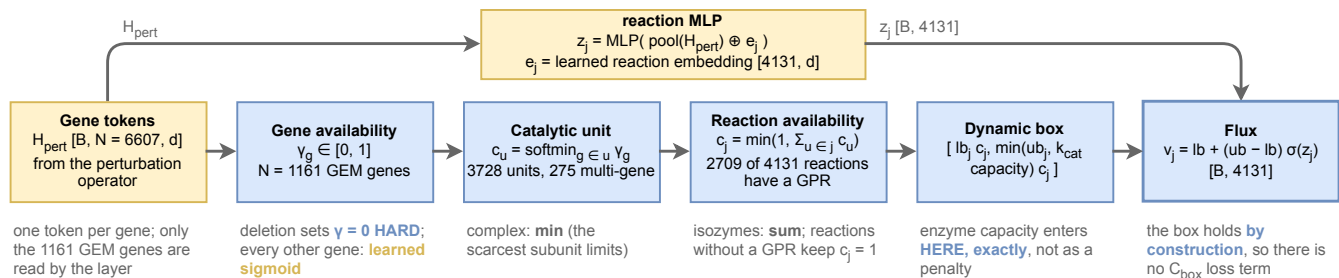
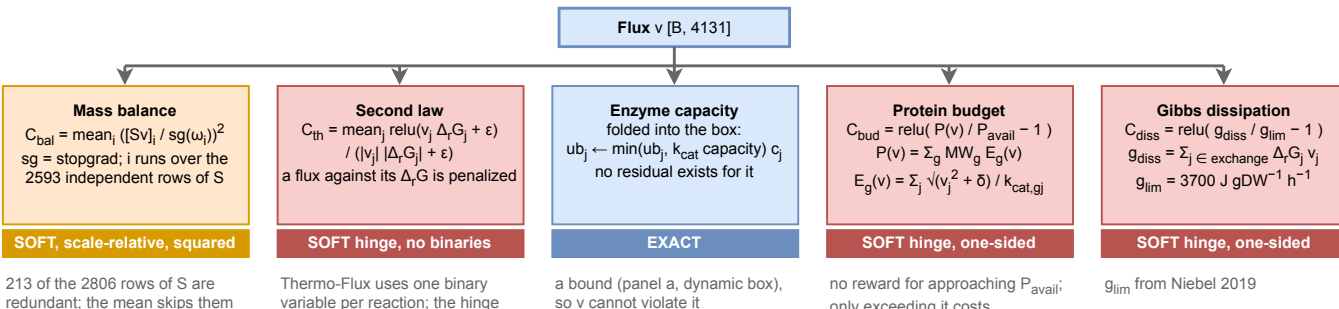


A differentiable enzyme-capacity constrained thermodynamic flux layer

a Availability to box to flux. A deletion acts on the bounds, so the box is exact and enzyme capacity never needs a penalty.



b The constraint residuals hanging off v. One is exact by construction; the other four are soft, and none needs a binary variable.



Thermodynamic decomposition of $\Delta_r G_j$, the driving force used by the second law and by dissipation

$$\Delta_r G_j = \Delta_r G_j^0 + RT \sum_i S_{ij} \ln c_i + \Delta_r G_j^{\text{t}} \quad \text{transport term: NOT AVAILABLE, set to 0}$$

tabulated: $\Delta_r G^0$ known for 85.1% of metabolites

learned $\ln c_i$, bounded to 1 μM to 10 mM; anchored where a measurement exists

What rests on data, and how much of it is there

S	100%
$\Delta_r G^0$	85.1% of metabolites
MW	100%
k_{cat}	4.0%: 148 of 3728 catalytic units resolved from the Open Enzyme Database
conc. anchors	6.8% of metabolites

